2.2 Classwork: The Cosine Rule — Markscheme

1. [Maximum mark: 7]

(a) $x^2 = 8^2 + 11^2 - 2(8)(11)\cos 40°$ **M1**

$x = 7.08 \quad (7.08351\ldots)$ **A1**

M1 for a correct substitution into the cosine rule. A value from a calculator in radian mode ($x = 17.4$) earns M1A0.

(b) The largest angle is opposite the longest side, 9.

$\cos\theta = \frac{5^2+7^2-9^2}{2(5)(7)}$ **M1**

$\cos\theta = -\frac{7}{70} = -0.1$

$\theta = 95.7° \quad (95.7391\ldots)$ **A1**

M1 for a correct substitution with 9^2 as the subtracted term. An angle found opposite the side 5 or 7 earns M0A0.

(c) obtuse, since $\theta > 90°$ **A1**

Accept "obtuse, since $\cos\theta < 0$". FT from (b).

(d) $y^2 = 6^2 + 10^2 - 2(6)(10)\cos 120°$ **M1**

$y^2 = 36 + 100 + 60$

$y^2 = 196$

$y = 14$ **A1**

The common error is a sign slip with $\cos 120° = -\frac{1}{2}$, giving $y^2 = 76$ and $y = 8.72$: M1A0.

2. [Maximum mark: 6]

(a) $d^2 = 6^2 + 9^2 - 2(6)(9)\cos 35°$ **M1**

$d = 5.34\,\text{km}$ $\quad (5.34149\ldots)$ **A1**

A diagram is not required.

(b) $AC^2 = 5^2 + 8^2 - 2(5)(8)\cos 60°$ **M1**

$AC^2 = 25 + 64 - 40$

$AC^2 = 49$

$AC = 7\,\text{cm}$ **A1**

(c) $\cos \angle ADC = \frac{4^2+6^2-7^2}{2(4)(6)}$ **M1**

$\cos \angle ADC = \frac{3}{48}$

$\angle ADC = 86.4°$ $\quad (86.4166\ldots)$ **A1**

M1 for a correct substitution with AC^2 as the subtracted term. FT from (b).